

Lecture 05: Training Versus Testing

Caltech CS156 Technical Study Guide

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September 18, 2026

Executive Summary

Lecture 05 formally initiates the Theory Track of CS156. It contrasts **Testing** (evaluating a single fixed hypothesis) with **Training** (searching over M hypotheses), explains why the naive Union Bound M fails for continuous hypothesis sets ($M = \infty$), introduces **Dichotomies** and the **Growth Function** $m_{\mathcal{H}}(N)$, defines the **Break Point** k , and proves the landmark result: if a break point exists, $m_{\mathcal{H}}(N)$ is strictly bounded by a polynomial in N , guaranteeing feasibility of learning.

1 Testing vs. Training: The Practice Exam Analogy

Consider preparing students for a final exam:

- **Testing (Validation):** A student is given a single fixed test h . Evaluating performance on a fresh set of questions measures true understanding ($E_{\text{out}} \approx E_{\text{test}}$) governed directly by Hoeffding's Inequality:

$$P(|E_{\text{test}}(h) - E_{\text{out}}(h)| > \epsilon) \leq 2e^{-2\epsilon^2 N} \quad (1)$$

- **Training (Learning):** A student is given practice problems with solutions. The student adjusts their hypothesis across M choices until E_{in} is minimal. Because h is chosen *after* seeing data, E_{in} is contaminated by data snooping/fitting.

2 Deconstructing the Loose Union Bound

To guarantee that $E_{\text{in}}(g) \approx E_{\text{out}}(g)$ for the selected hypothesis $g \in \mathcal{H}$, we bound the probability of any "bad event" $B_m = [|E_{\text{in}}(h_m) - E_{\text{out}}(h_m)| > \epsilon]$:

$$P\left(\bigcup_{m=1}^M B_m\right) \leq \sum_{m=1}^M P(B_m) \leq 2Me^{-2\epsilon^2 N} \quad (2)$$

The Union Bound assumes worst-case disjoint bad events. In reality, bad events for similar hypotheses (e.g., two perceptrons separated by a tiny angle) are **heavily overlapping**. Charging M times overcounts redundant hypotheses. For continuous models (where $M = \infty$), the Union Bound is infinite and useless.

3 Dichotomies & The Growth Function

To fix the Union Bound, we count hypotheses **restricted to a finite sample** $\mathbf{x}_1, \dots, \mathbf{x}_N$.

3.1 Dichotomies

A **dichotomy** is a hypothesis restricted to N points:

$$h : \{\mathbf{x}_1, \dots, \mathbf{x}_N\} \rightarrow \{-1, +1\} \quad (3)$$

While $\mathcal{H}$ may contain infinitely many functions on $\mathcal{X}$, it can generate at most 2^N distinct binary label patterns (dichotomies) on N points.

3.2 Definition of Growth Function $m_{\mathcal{H}}(N)$

The **Growth Function** $m_{\mathcal{H}}(N)$ is the maximum number of dichotomies generated by $\mathcal{H}$ on any N points:

$$m_{\mathcal{H}}(N) = \max_{\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathcal{X}} |\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_N)| \leq 2^N \quad (4)$$

4 Growth Functions of Standard Models

1. **Positive Rays** ($d = 1$): $h(x) = \text{sign}(x - a)$. Placing N points creates $N + 1$ possible dichotomy splits:

$$m_{\mathcal{H}}(N) = N + 1 \quad (\text{Linear Growth}) \quad (5)$$

2. **Positive Intervals** ($d = 1$): $h(x) = +1$ inside $[a, b]$, -1 outside. Choosing 2 endpoints among $N + 1$ segments yields:

$$m_{\mathcal{H}}(N) = \binom{N+1}{2} + 1 = \frac{1}{2}N^2 + \frac{1}{2}N + 1 \quad (\text{Quadratic Growth}) \quad (6)$$

3. **Convex Sets** ($d = 2$): Any convex shape. Placing N points on the perimeter of a circle allows any subset of points to be assigned $+1$ via a polygon connecting them:

$$m_{\mathcal{H}}(N) = 2^N \quad (\text{Exponential Growth - Shatters All } N) \quad (7)$$

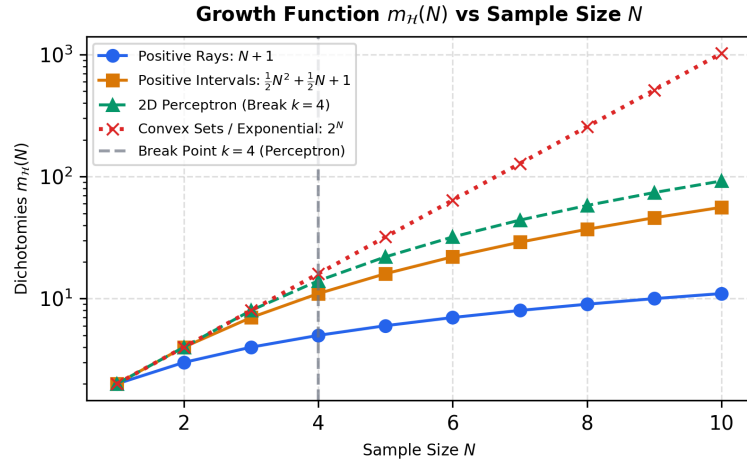


Figure 1: Growth Functions $m_{\mathcal{H}}(N)$ for Various Models.

5 Break Points & Polynomial Bounding

When $m_{\mathcal{H}}(N) = 2^N$, $\mathcal{H}$ **shatters** the N points (can implement all 2^N dichotomies).

5.1 Definition of Break Point k

If no dataset of size k can be shattered by $\mathcal{H}$, then k is a **Break Point** for $\mathcal{H}$:

$$m_{\mathcal{H}}(k) < 2^k \quad (8)$$

For example, a 2D Perceptron can shatter any 3 points ($m_{\mathcal{H}}(3) = 8 = 2^3$), but fails on non-separable 4-point patterns (e.g., XOR), achieving only $m_{\mathcal{H}}(4) = 14 < 16$. Thus, $k = 4$ is the break point for 2D Perceptrons.

5.2 The Fundamental Theorem

Main Result of Learning Theory

If a hypothesis set $\mathcal{H}$ has a break point k , then for all N :

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^{k-1} \binom{N}{i} = \mathcal{O}(N^{k-1}) \quad (9)$$

Takeaway: If a break point exists, the growth function collapses from exponential 2^N to a polynomial $\mathcal{O}(N^{k-1})$. In Hoeffding's generalization bound, a polynomial growth function is completely dominated by the negative exponential $e^{-2\epsilon^2 N}$, proving that **learning is feasible for infinite hypothesis spaces!**

6 Combinatorial Puzzle: Break Point $k = 2, N = 3$

To illustrate how a break point severely restricts dichotomies:

- Sample size $N = 3 \implies 2^3 = 8$ possible dichotomies.
- Break point $k = 2 \implies$ no pair of points can display all 4 binary patterns (00, 01, 10, 11).

Attempting to construct a 3-column binary table without violating $k = 2$ reveals that at most ****4 dichotomies**** can exist (a reduction of 50% from 8).